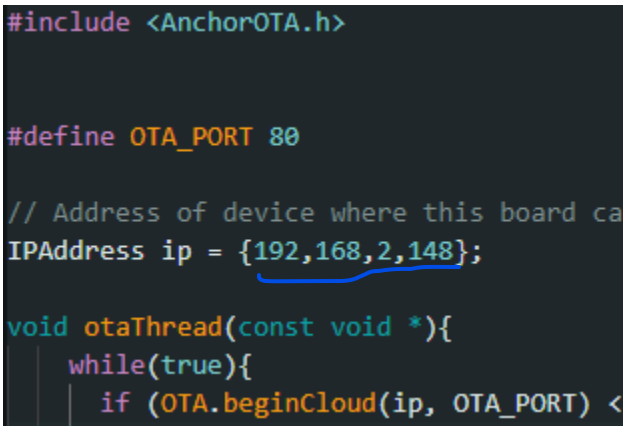
FLASHING USING LOCAL PYTHON OTA SERVER: (TEMPORARY SOLUTION)

SETUP:

For Client: (person in possession of voltlink)

1. Change the local IP address in the AnchorhttpOTA.h file in the firmware to the developers(the person that will use the OTA for flashing) local IPV4 address.**(very important)**



2. Compile the code .
3. Navigate to the folder where the binary is present :

   C:\Users\eSpark\AppData\Local\arduino\sketches\BBE5FD9EA76ED39DB89B56664D98DE06

   should be something like this. The last folders name will be different , you can find it by filtering the folders to the latest modified .

4. Copy the “km0_km4_image2.bin file” which is the main binary file, this binary will contain the functionality for the local python OTA server
5. The person which is going to use the OTA flashing method will require the board flashed with this binary and the python Download Server.

For Developer:(without voltlink)

1. After having the board in possession with the firmware containing the OTA functionality
2. Setup the python Download Server (may or may not require to install dependencies)

3. In the DownloadServer.py change the IP address to your local IPV4 address the binary file name can also be changed if needed.

```
import os

int_max_value = 4294967295

filename = "km0_km4_image2.bin"
fixed_ip = "192.168.2.148"  # Set your fixed IP address here
port = 80

f = open(filename, "rb")
checksum = 0
length = os.stat(filename).st_size
```

4. Place the binary file “km0_km4_image2.bin” containing OTA functionality in the same directory as the Download Server.

How to Flash? :

- Now if the setup is correct and both the board and python server are on the same IP of the Developer Everything should work as intended.
- Power up the board
- The board will start to listen for the Binary file.
- Now in the terminal navigate to the directory where DownloadServer.py is present and run the following command:

```
ies\AnchorOTA\DownloadServer> python DownloadServer.py
```

On successful flash following logs will be shown in the terminal:

```
Checksum: 0x5c0a12d
Length: 966656
Listening on 192.168.2.148:80
Connected by ('192.168.2.30', 64397)
Sent header
Sent binary
```

and the firmware will be updated.

- Make sure to kill the terminal where the DownloadServer is running in about 12-13 seconds after running the command or else it will keep flashing in loop.